



Optimizing high-strength concrete compressive strength with explainable machine learning

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Abstract

This study leverages machine learning to enhance the prediction of high-strength concrete (HSC) compressive strength, addressing the limitations of conventional methods, which are often tedious, less reliable, and time-consuming. Extreme Gradient Boosting (XGB) serves as the primary model, with hyperparameter optimization via metaheuristic algorithms such as Cuckoo Search (CSA), Water Strider (WS), Leopard Seal (LS), Harris Hawk (HH), Invasive Weed (IW), and Forest Optimization (FO). A total of 681 data sets were collected from existing literature. The models underwent tenfold cross-validation, with the LS-XGB model achieving an almost ideal performance in testing sets. Other models, including CSA-XGB, WS-XGB, HH-XGB, IW-XGB, and FO-XGB, also demonstrated strong performance, each with $R^2 > 0.96$. For model explainability, Shapley's Additive Explanation (SHAP) analysis has been applied to the best-performing LS-XGB model. The analysis revealed that cement and superplasticizer (SP) are the most crucial features contributing to HSC development, with optimal ranges identified at 600–900 kg/m³ for cement and 8–10 kg/m³ for SP. The study demonstrates on how feature interactions contribute to concrete materials compressive strength, providing better and above all sustainable constructions. Furthermore, the LS-XGB model's optimal performance depicts the strongly nonlinear nature of HSC materials, validated through a set of derived graphs. Additionally, 30 concrete cubes were prepared for experimental validation, and the datasets demonstrated an accuracy of 92% showcasing the ability of models to make well informed decision.

Keywords High-strength concrete · Ensemble learning · SHAP analysis · Metaheuristics algorithm · Nature inspired algorithm

1 Introduction

Cement-based concrete is widely used in civil engineering due to its strength which is the key factor for special structures such as tall buildings, prefabricated structures, dams, bridges, etc. (Tejas and Pasla 2023). HSC reduces the size of bearing structural elements, decreases reinforcement and formwork needs, minimizes creep and shrinkage, and increases stability and lifespan under various loading conditions (Naik et al. 2012; Neupane et al. 2018). Designing HSC is more complex than regular concrete, requiring more technical expertise and a deep understanding of its properties (Abbas et al. 2024). The Key factors in HSC development include cement, water, aggregates, and superplasticizers. Cement amounts depend on strength, workability, and site conditions (Ayanlere et al. 2023). A low water-cement (W/C) ratio is crucial for low permeability and high durability (Zeyad et al. 2020). The proper amount of fine aggregates (FA)

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increase strength, while coarse aggregates (CA) improve effective porosity and mechanical strength (Donza et al. 2002; Yuan et al. 2023). Furthermore, superplasticizers significantly increase compressive strength especially for small values of water-to-binder ratio. The other important factors include environmental conditions, workability, and aggregate size (Alsayed 1998; Houst et al. 2008; Park et al. 1999; Shah and Ribakov 2011).

Concrete possesses a highly non-linear relationship with its components. Thus, the assessment of CS of concrete demands necessitates the conduction of various mechanical tests for accurate evaluation. However, on-site testing is highly labor-intensive and requires considerable human effort, leading to the possibility of occurrence of several technical errors (Jacobsen et al. 1995; Sapkota et al. 2023). Subsequently, estimating the CS of concrete in a laboratory is tedious and demands exploring new and more convenient methods. Numerous researchers used statistical and regression models to estimate concrete CS to mitigate such inconveniences (Alyaseen et al. 2024; Alyousef et al. 2023; Asadi Shamsabadi et al. 2022; Naseri et al. 2020; Sun et al. 2022). Such models need to comprehend the complexity of concrete constituents and CS, resulting in relatively lower accuracy. This led to the emergence of machine learning (ML) methods that unraveled traditional methods' limitations and gave new insights into establishing a more accurate prediction model. Moreover, such techniques performed remarkably well in predicting the mechanical properties of the concrete with less computational time and limited resources.

With the rise of Machine learning, several civil engineering domains have been adopted to address the existing problem, including concrete science (Karim et al. 2024; Saha et al. 2024; Y. Yang et al. 2024a), geotechnical (Ghani et al. 2024; Kumar et al. 2022), structural (Kaveh and Khavaninzadeh 2023; Mangalathu and Jeon 2018; Sapkota et al. 2024a, b), and many more (Almubaidin et al. 2024; Mihel et al. 2024; Son et al. 2012). Such data-driven methods are capable of capturing complex nonlinear relationships among the variables. Several past studies have successfully demonstrated the use of ML for better concrete assessment. Since HSC developments witnessed several uncertainties in the design mix, ML models can be an alternative for predicting mechanical properties. For instance, Miao et al. (2024) used ML models like Back propagation neural network (BPNN), Extreme Gradient Boosting (XGB), Support vector regression (SVR), and Random forest regressor (RFR). They further used the Sparrow search algorithm (SSA) to develop four hybrid models (SSA-BPNN, SSA-XGB, SSA-SVR, SSA-RFR) to predict the CS of glass powder concrete. The study revealed that SSA-XGB was the superior model with $R^2 = 0.9645$ and $RMSE = 3.0640$ MPa. Ahmad et al. (2023) used linear regression and ANN model using polypropylene

fiber to predict the CS of HSC. The ANN model provided the best results with $R^2 = 0.96$ and $RMSE = 4.02$ MPa. Likewise, Kumar Tipu et al. (2022) incorporated recycled coarse aggregates and Newton's boosted backpropagation neural network (NB-BPNN) to predict the CS of concrete. The result demonstrated that the NP-BPNN model exhibited impressive results with $R^2 = 0.95$ and $RMSE = 3.205$ MPa. Thi Mai et al. (2023) used the ML approach to predict the CS of Roller Compacted Concrete (RCC). They found that the CatBoost model was the most efficient, exhibiting $R^2 = 0.983$. Furthermore, they emphasized the effects of cement concrete and age on the CS, which help civil engineers optimize the composition of RCC. Similarly, Zhao et al. (2024) used five optimization algorithms, namely the slime mold algorithm (SMA), black hole algorithm (BHA), multi-verse optimizer (MVO), Vortex Search (VS), and whale optimization algorithm (WOA) to analyze the compressive strength of concrete. This study revealed that the VS predicted the most accurate result with a statistical R^2 of 0.9841 in the training phase and 0.9888 in the testing phase. Additionally, Yang et al. (2024b) employed multiple ML algorithms, including genetic programming (GP), support vector regression (SVR), XGB, Back Propagation artificial neural network (BP-ANN), and adaptive network-based fuzzy inference system (ANFIS) and particle swarm optimization (PSO) to predict the CS of environmentally friendly concrete. The result showed that the PSO-XGB model exhibited the most accuracy with $R^2 = 0.9072$. Maherian et al. (2023) used the ML approach to estimate the CS of nano-silica-modified concrete. They found that among various ML models, the Random Forest (RF) method and the artificial neural network (ANN) models were most efficient with R^2 values of 0.93 and 0.92, respectively. Beskopylny et al. (2024) used ML methods like linear regression (LR), Support Vector Regression (SVR), Random Forest (RF), and CatBoost to predict the CS of vibro-centrifuged concrete. The study revealed that the CatBoost model was the most efficient, with $R^2 = 0.94$. Furthermore, Sun et al. (2024) adopted hybrid ML techniques to investigate the CS of coral aggregate concrete (CAC). The result showed that the genetic algorithm-optimized backpropagation neural network (GA-PB) model provided the best result with correlation coefficient (R) = 0.98 and $RMSE = 3.33$. This study helped provide practical guidelines for CAC preparation. Khan and Abbas (2023) used a multilayer stack model to predict the CS of reactive powder concrete (RPC). The result demonstrated the stacking model to be the most optimal with $R^2 = 0.96$. This study showed a strong relationship between the strength of RPC and its corresponding mixture elements. Chen (2024) used XGB and bagging regressor (BR) to estimate marble cement mortar's compressive and flexural strength. The result showed that XGB with $R^2 = 0.98$ outperformed the BR model with $R^2 = 0.94$. Furthermore, Wang et al. (2024) employed ML techniques like decision

Table 1 List of ML models used for assessing the CS of concrete

Study	Dataset	Model used	Findings/Results
Mai et al. (2023)	387	XGB, DT, LGBM	$R^2 = 0.992$, RMSE = 1.892 MPa by XGB
Kashem et al. (2024a, b)	810	RF, AB, GBM with PSO	$R^2 = 0.9804$, RMSE = 5.95 MPa in the testing stage with hybrid GBM-PSO model
Katlav et al. (2024)	181	RF, XGB, LGBM and AB	$R^2 = 0.948$, RMSE = 6.68 MPa in the test phase
Katlav and Ergen (2024)	785	PPSO, DMO, ASO with CB	$R^2 = 0.978$, RMSE = 6.15 MPa in the testing stage with hybrid DMO-CB model
Asteris et al. (2021)	1030	HENSM	$R^2 = 0.8894$, RMSE = 0.0701 MPa in testing set
Demetriou et al. (2024)	1030	HCR	$R^2 = 0.90$, RMSE = 5.26 MPa in testing set
Kashem et al. (2024a, b)	1404	LGBM, XGB, RF	$R^2 = 0.95$, and RMSE = 5.255 MPa by XGB-LGBM
Peng and Unluer (2023)	607	ANN, SVM, GWO	$R^2 = 0.9056$, RMSE = 4.549 MPa by hybrid GWO-SVR model
Paudel et al. (2023)	633	MLR, SVR, AB, RF, XGB, BR	R^2 of 0.95, RMSE = 3.06 MPa by XGB
Hu et al. (2023)	351	BPNN, AB, SVM, KNN, RBFNN	R of 0.9921, RMSE = 15.4 MPa by BPNN
Pakzad et al. (2023)	176	CNN, KNN	R^2 of 0.928, RMSE = 5.043 MPa by CNN

DT Decision Tree, LGBM Light Gradient Boosting Method, RF Random Forest, AB AdaBoost, GBM Gradient Boosting Method, PSO Particle Swarm Optimization, SVM Support Vector Machine, ANN Artificial Neural Network, SVR Support Vector Regression, GWO Grey Wolf Optimization, GA Genetic Algorithm, CBO Colliding Bodies Optimization, ECBO Enhanced Colliding Body Optimization, MLR Multiple Linear Regressor, BR Bagging Regressor, BPNN Back Propagation Neural Network, KNN K-Nearest Neighbor, RBFNN Radial Basis Function Neural Network, CNN Convolutional Neural Network, LR Linear Regression, HENSM Hybrid Ensemble Model, ANN Artificial Neural Networks, HCR Hierarchical Classification and Regression, PPSO Phasor particle swarm optimization, DMO Dwarf mongoose optimization, ASO atom search optimization, CB CatBoost

tree (DT), random forest regressor (RFR), AdaBoost regressor (AR), and gradient boosting regressor (GBR) to predict the CS of self-compacting concrete (SCC). The study showed AR and RFR models with R^2 values of 0.90 and 0.91, respectively, were superior to DT and GBR models with R^2 values of 0.85 and 0.88, respectively. Aslam and Zubair (2024) used eleven ML models, namely gene expression programming (GEP), ANN, RFR, SVR, ANFIS, optimized Gaussian process regressor (OGPR), gradient boosting (GB), XGB, AdaBoost, k-nearest neighbor (KNN), and BR to predict the CS of blended cement concrete (BCC). The result of the study demonstrated that XGB was the most accurate model with $R^2 = 0.89$. (Javed et al. (2024a, b) incorporated SVR with three metaheuristics optimization algorithms, namely PSO, grey wolf optimization (GWO), and firefly algorithm (FFA), for the prediction of CS of steel fiber reinforced concrete when exposed to high temperatures. The study revealed that the SVR-FFA and SVR-GWO hybrid models exhibited high accuracy with an R-value exceeding 0.99 for both models. Sun and Lee (2024) utilized natural gradient boosting (NGB) to analyze the CS of oil palm shell (OPS) -based lightweight aggregate concrete (LWAC). Their analysis revealed that the NGB model demonstrated a highly accurate performance with an R^2 value of 0.969 in the testing phase. Alyami et al. (2024) used seven ML algorithms, namely SVR, DT, SVR-Bagging, SVR-Boosting, RF, GB, and GEP, for predicting the CS of 3D printed fiber-reinforced concrete. The result

of the study revealed that DT, RF, GB, and GEP models exhibited substantial accuracy with R values of 0.987, 0.986, 0.986, and 0.98, respectively. Table 1 summarizes some literature on using ML to assess concrete's CS. The literature shows that ML models have the potential to predict concrete CS more accurately. Further, it shows that advanced ensemble techniques have surpassed traditional algorithms with minimal errors. This study applied a hybrid model that incorporates nature-inspired metaheuristic algorithms to improve the predictability of concrete compressive strength. The model is validated with laboratory experiments. This research addresses challenges in optimizing concrete mix designs, particularly in developing countries like Nepal and India, to reduce construction cost and time. The critical gaps and significance of the study is presented in Sect. 2. Likewise, the adopted ML model and optimization techniques are explained in Sect. 3, the data and statistical methods are presented in Sect. 4, the experimental methodology is described in Sect. 5, model building and tuning is covered in Sect. 6, performance is assessed in Sect. 7, the results are discussed in Sect. 8, and the conclusion includes important findings and future implications.

2 Research significance

The significance of this study lies in its novel approach to predicting the compressive strength of high-strength concrete using hybrid machine learning. Recent works by Kashem and Das (2023), Das et al. (2024), and Davawala et al. (2023) have demonstrated the utility of ML techniques like XGBoost, Random Forest, Gradient Boosting and support vector regressions for predicting the CS of various types of concrete. Despite sustainable materials being an excellent alternative to existing construction materials, they have not been extensively adopted in real-time construction scenarios. Thus, optimizing the currently adopted materials in construction can address on-site challenges. Modifications to the existing mix design, including increasing the amount of cement, reducing the W/C ratio, and using superplasticizers, can develop HSC from ordinary concrete mix. Since existing literature integrates different data sets from numerous studies to assess the mechanical properties of concrete using ML techniques, the robustness of the model relies on the experimental data used for validation. The studies mentioned in the literature focused on utilizing standalone and ensemble model techniques (Ergen and Katlav 2024). However, such models showed compromised performance, leading to the adoption of hybridization in ML. The trend of adopting such methodologies has been widely used. However, the problems associated with such techniques are the model's explainability behavior and accuracy. Thus, adopting such robust and explainable techniques helps researchers and engineers make well-informed decisions.

The research outperforms any previous ML studies by developing a LS-XGB hybrid model where XGBoost learning is integrated with nature-inspired metaheuristic algorithms (NIMAs). Moreover, the improved transparencies gained through using SHAP-based analyses, extends beyond regular performance metrics, providing insight into the contribution of each input features and therefore allowing for more informed decisions by researchers and practitioners.

This is particularly important for field applications of construction, because design changes through increasing the water-to-cement ratio, higher cement amount, as well as utilizing superplasticizers, can affect the conventional concrete mixtures in the design required. Previous studies have demonstrated the potential for predicting concrete strength using such big, aggregated datasets, but their models may not achieve the same accuracy when validated on smaller, pragmatic datasets, which do not share resultant variability or constituent proportions. To fill this gap, the current study is not only showing how effective their model is on already published data but also validating these results with data from an experimental lab. This strategy not only assures the reliability of the suggested hybridized model but also its practical feasibility.

Additionally, the research emphasizes the significance of explainable machine learning in civil engineering settings. Most previous research on ensemble and hybrid models have emphasized on the predictive accuracy improvement over using a single model and thus have overlooked the aspect of interpretability as well. Using the SHAP-based analyses, this work provides transparent insights into the extent to which each feature (cement content, water-to-cement ratio, and dosages of other admixtures) contribute to the predicted compressive strength. Such clarity is needed to resolve the real-world constraints that engineers encounter on construction sites. In conclusion, the proposed approach puts forward a possibility for incorporation of advanced data-driven methodologies in standard concrete mix design, greatly enhancing the predictive accuracy and informed decision-making of current concrete construction projects. Such methods assist in optimizing the usage of cement in several massive projects in developing countries like Nepal, India, where advanced construction projects are yet to expand. This research elucidates regular construction practices and observes trends within the field. Enhancing the efficient use of concrete components can yield considerable reductions in time and cost, especially in projects with high concrete usage. The schematic working diagram of this study is presented in Fig. 1.

3 Machine learning techniques

3.1 Extreme Gradient Boosting (XGB)

XGB is a robust method based on the gradient-boosting framework for regression and classification issues (Nguyen et al. 2022). It is used to handle many intricate and non-linear interactions within the data. XGB operates with a parallel tree boost acquiring superior results in numerous ML problems. XGB gives optimal output by combining predictions from multiple decision trees. Gradient boosting seeks to train the decision trees so that the succeeding trees correct the mistakes made by preceding trees. Gradient boosting contains three basic steps: reducing the loss function, using a weak learner for prediction, and including an additive model for loss function restriction (Chen and Guestrin 2016). XGB iteratively adjusts the parameters by leveraging residuals from the previous phase to optimize the goal function. Various hyperparameters, like the learning rate, number of trees, and regularization, influence the performance of XGB. Figure 2 displays the schematic representation of the XGB model. The loss function can be calculated as in Eq. (1):

$$\mathcal{L}^t = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (1)$$

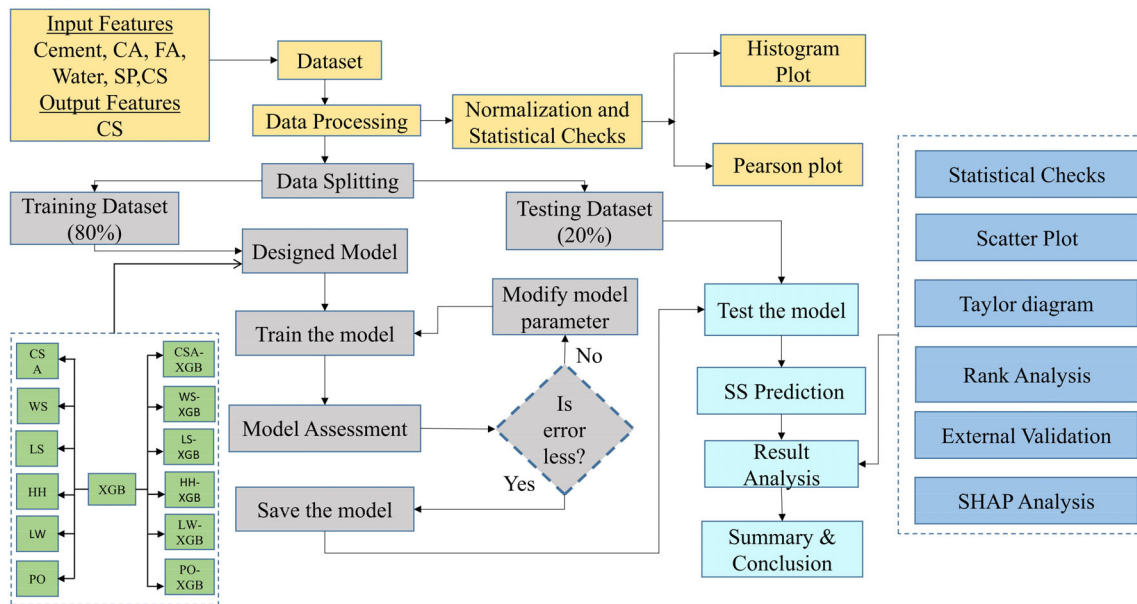


Fig. 1 Schematic diagram of the study

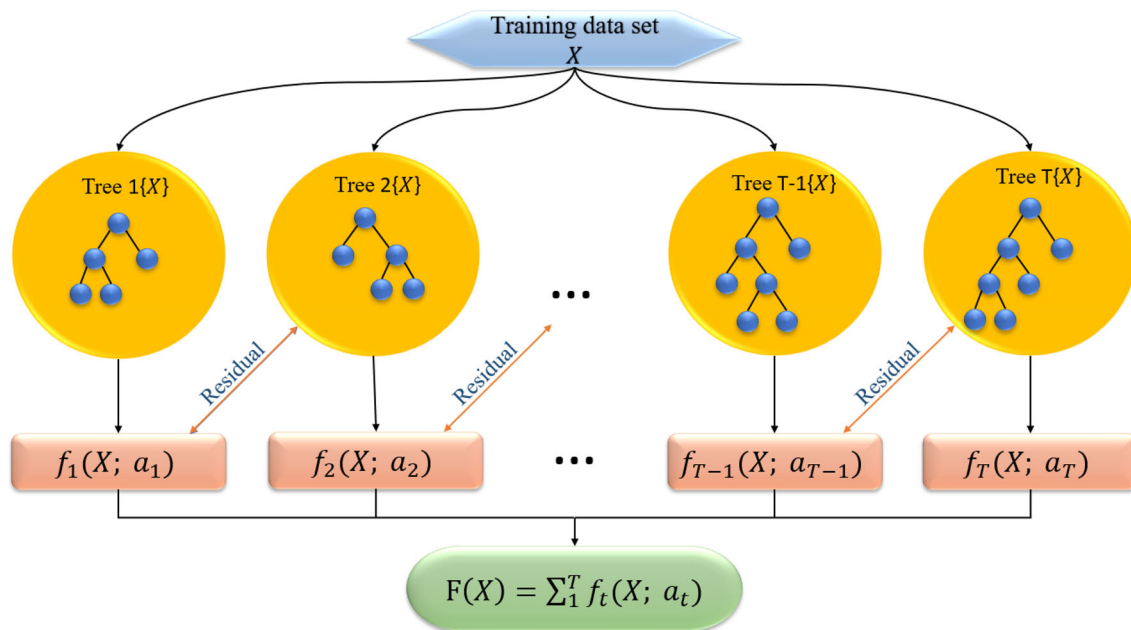


Fig. 2 Working of XGB model

where, l is loss function between predicted and observed data, $y_i, \hat{y} =$ observed data and predicted data respectively, $f_t =$ model of t^{th} tree, $t =$ iteration index, $\Omega(f_t)$ is given by Eq. (2):

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (2)$$

where, T is the total number of tree leaves λ and γ are penalty coefficient, ω is vector having score of each leaf.

3.2 Metaheuristic algorithm

Meta-heuristic algorithms (MHA) are techniques that are developed to optimize a solution to a convoluted problem. These algorithms are much more adept at finding quality solutions to problems in less time. MHA has two essential features, namely exploration and exploitation, which indicate the algorithm's capability to search for new spaces and find the optimal solution in favorable search spaces, respectively. Numerous sources, including evolutionary techniques,

swarm intelligence, and physics-based and human-based optimization techniques for optimal fitness function inspire MHA. Some of the new MHA inspired by animals and plant behavior include ant lion (Ehteram et al. 2023), whale (Mirjalili and Lewis 2016), leopard seal (Rabie et al. 2023), cuckoo (Liu et al. 2023; Mareli and Twala 2018), Harris hawk (Heidari et al. 2019), forest (Ghaemi and Feizi-Derakhshi 2014), Invasive weed (Kumar et al. 2020), and many more (Gandomi and Alavi 2012; Kaveh and Dadras Eslamlou 2020; Mirjalili 2016; Mirjalili et al. 2014; Yang and Gandomi 2012).

3.2.1 Cuckoo Search Algorithm

Among the numerous nature-based optimization techniques that have gained extensive attention among researchers, Cuckoo search optimization algorithm is the one that uses the brood reproductive strategy nature of Cuckoo birds. The strategy involves laying their eggs in the nests of other birds. When other birds discover the cuckoo eggs, they will either get rid of those eggs or desert their nest and build a new one. This tendency of other birds affects the survival rate of cuckoo in the upcoming generation.

The CSA algorithm performs its search process in two phases: global search and local search (Liu et al. 2023). For global search, it uses Lévy flights, a behavioral theory used to replicate the cuckoo's reproduction practices closely. Lévy flights are random walks whose step is obtained from Lévy distribution (Mareli and Twala 2018). For local search, a biased random walk is used. The new position can be calculated from Eqs. (3–5):

When a cuckoo lays one egg randomly in a host bird's nest, the probability of finding that egg by the host bird is P_a ($P_a \in [0, 1]$). Let us assume N number of cuckoos, the position of the m^{th} cuckoo is denoted by Z_i and t is the ongoing iteration. Let s be the step size of the Lévy flight.

$$z_m^{t+1} = z_m^t + \alpha \otimes \text{Lévy}(s, \lambda) \quad (3)$$

where, α = step size and the product, $\otimes$ denotes the multiplication of the corresponding position of the vector.

$$\text{Lévy}(s, \lambda) = \frac{\lambda \Gamma(\lambda) \cdot \sin\left(\frac{\pi\lambda}{2}\right)}{\pi} \cdot \frac{1}{s^{1+\lambda}} \quad (4)$$

$$\sigma_\mu = \left\{ \frac{\lambda \Gamma(1+\lambda) \cdot \sin\left(\frac{\pi\lambda}{2}\right)}{\Gamma\left[\frac{(1+\lambda)}{2}\right] \cdot \lambda \cdot 2^{\left(\frac{\lambda-1}{2}\right)}} \right\}^{\frac{1}{\lambda}} \quad (5)$$

3.2.1.1 Water Strider Algorithm Water Strider Algorithm, inspired by the life cycle of water strider bugs, is a population-based optimizer. The water Strider Algorithm comprises the same functions as those exhibited by the water strider bugs,

like territorial behavior, intelligent ripple communication, mating style, feeding mechanism, and succession (Kaveh and Dadras Eslamlou 2020). The essential step of the Water Strider Algorithm involves initial birth, followed by territory establishment, mating feeding and ending with death. The equation engaged in the water strider algorithm is equated in Eqs. (6–8)

$$WS_i^0 = Ub + \text{rand}(Ub - Lb), i = 1, 2, \dots, nws \quad (6)$$

where, WS_i^0 = initial position of i^{th} water striders, Ub and Lb are upper and lower limit of values, rand is a random constant number between 0 and 1, inclusive, and nws is the number of WS_s . Likewise, for the total number of nt territories, WS are arranged according to their fitness by making $\frac{nws}{nt}$ number of groups. The j^{th} member of each group is assigned to j^{th} territory. The position in each territory is considered as the male and female with worst and best fitness, respectively. For mating, male WS sends ripple to female WS . A probability p is assumed for attraction and $1-p$ for repulse response.

$$\begin{cases} WS_i^{t+1} = WS_i^t + R \cdot \text{rand}; & \text{if mating happens with probability } p \\ WS_i^{t+1} = WS_i^t + R \cdot \text{rand}; & \text{otherwise} \end{cases} \quad (7)$$

$$R = WS_F^{t-1} - WS_i^{t-1} \quad (8)$$

where, WS_F^{t-1} and WS_i^{t-1} = male and female WS in $(t-1)^{\text{th}}$ cycle. Similarly, Male WS look after food as mating consumes lot of energy for water striders. To know the availability of food, objective function is computed. If computed value of objective function is higher than the previous, the male WS has found recovery food, otherwise not. If It has not found food, the male WS must move to the place having highest fitness (WS_{BL}^t) according to the formula shown in Eq. (9):

$$WS_i^{t+1} = WS_i^t + 2\text{rand}(WS_{BL}^t - WS_i^t) \quad (9)$$

If the male WS becomes unable to find food, it will terminate and a new WS will replace it as shown in Eq. (10):

$$WS_i^{t+1} = Lb_j^t + 2\text{rand}(Ub_j^t - Lb_j^t) \quad (10)$$

Lb_j^t and Ub_j^t are the maximum and minimum boundaries of WS in j^{th} territory. Finally, If termination condition is satisfied, the algorithm stops and gives the result, otherwise algorithm returns to the mating step until the termination condition is not satisfied.

3.2.1.2 Leopard Seal Algorithm Leopard Seal Optimization (LSO) is a nature-inspired optimization algorithm inspired by the hunting nature of leopard seals. Leopard seals are always in the search of their prey even if they are not hungry (Rabie et al. 2023). Their behavior shows that they distribute themselves so that each individual is tasked with searching a specific range. When one of the seals finds prey, the herd members near the area are summoned, and they attack together.

LSO is modelled into three sequential phases: (I) Searching for prey (II) Encircling the prey and (III) Attacking the prey. These phases depict the dominant predating nature of the lion seals and are mathematically expressed as equated in Eqs. (11–16).

$$\vec{X}_{\alpha}(L_m) | \forall \alpha \in 1, 2, 3, \dots, \\ \xi - 1 = \vec{D}^i(L_m) \cdot e^{bf_{\alpha}} \cos(2\pi f_{\alpha}) + \vec{X}_{\xi}^i L_m \quad (11)$$

where $\vec{X}_{\alpha}^i(L_m)$ is the updated α 's position of L_m at i^{th} iteration and $\vec{D}^i(L_m)$ is the distance between leopard seal and prey.

$$\vec{D}^i(L_m) = \left| \vec{X}_{\xi}^i(L_m) - \vec{X}_1^i(L_m) \right| \quad (12)$$

$$f_{\alpha} = 1 - \frac{2 \times \alpha}{\xi} \quad (13)$$

Equation (17) can be used to find the total number of allowed iterations (N) as:

$$N = N_{Srch} + N_{Enc} + N_{Att} \quad (14)$$

where, N_{Srch} , N_{Enc} , N_{Att} are the iterations for searching, encircling, and attacking respectively.

$$\vec{X}_{prey} = \frac{1}{k} \begin{pmatrix} \sum_{i=1}^k x_{1i} \\ \sum_{i=1}^k x_{2i} \\ \dots \\ 0pt \\ \dots \\ \sum_{i=1}^k x_{zi} \end{pmatrix} \quad (15)$$

$$\vec{X}_{prey} = \frac{1}{\sum_{m=1}^k W(L_m)} \begin{pmatrix} \sum_{m=1}^k W(L_m) \cdot x_{1m} \\ \sum_{m=1}^k W(L_m) \cdot x_{2m} \\ \dots \\ 0pt \\ \dots \\ \sum_{m=1}^k W(L_m) \cdot x_{zm} \end{pmatrix} \quad (16)$$

3.2.1.3 Harris Hawk Algorithm The Harris Hawks (HHO) algorithm is a population-based, metaheuristics optimization approach. This approach is inspired by how the Harris Hawks hunt their prey using “surprise pounce.” The hawks converge on a running rabbit from several directions, so the prey is surprised (Heidari et al. 2019). This tracing and circumscribing approach to attacking inspires the HHO algorithm. The expression of the Harris hawk is equated on Eq. (17–20).

$$Q(k+1) = \begin{cases} Q_{rand}(k) - f_1 |Q_{rand}(k) - 2f_2 Q(k)|, & b \geq 0.5 \\ (Q_r(k) - Q_{mean}(k)) - f_3(L_B + f_4(L_B - U_B)), & b < 0.5 \end{cases} \quad (17)$$

where, $Q(k)$ and $Q(k+1)$ are the locations of hawk in the current and upcoming iteration, respectively. f_1, f_2, f_3, f_4 and b are random numbers within (0,1). $Q_i(k)$ be the location of i^{th} Harris Hawk in iteration k and N be the total number of hawks. Then, Eq. (18) is used for calculating the mean position $Q_{mean}(k)$ while Levy is calculated from Eq. (20)

$$Q_{mean}(k) = \frac{1}{N} \sum_{i=1}^N Q_i(k) \quad (18)$$

$$Z = 2Z_o \left(1 - \frac{k}{k_{max}} \right) \quad (19)$$

$$Levy(x) = 0.01 \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}}, \sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right)^{\frac{1}{\beta}} \quad (20)$$

where, u, v are random values within (0,1) and β is constant equals 1, Y is the random vector and D is the dimension.

3.2.1.4 Invasive Weed Algorithm Invasive weed optimization algorithm is influenced by the natural phenomena of weeds in the agricultural fields. Weeds are the unwanted things on the field that affect cultivated plants' growth and yield capacity due to its vigorous expansive phenomena. The principle of invasive weed optimization initially involves where each seed produces plants and produces new seeds. The spatial distribution of weeds, a natural agency, then disperses the produced seeds throughout the field. Finally, the cycle terminates when the number of plants reaches the maximum limit in the field, and the more potent or best-suited plants survive and produce new seeds. All the unsuitable seeds are disregarded, which is termed competitive elimination.

The following Eq. (21) is used to store the set of population X : This matrix yields the population set for a D -dimensional

problem with a population size of N . The lower and upper bounds L and U having length D can be presented as Eq. (22):

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1D} \\ x_{21} & x_{22} & \dots & x_{2D} \\ \dots & \dots & \dots & \dots \\ x_{N1} & x_{N1} & \dots & x_{ND} \end{bmatrix} \quad (21)$$

$$L = [L1L2 \dots LD] \quad (22)$$

$$D = [U1U2 \dots UD]$$

The number of seeds produced by a seed can be evaluated with Eq. (23), which is given by:

$$S_p = S_{min} + (F_p - F_w) \left\{ \frac{F_p - F_w}{S_{max} - S_{min}} \right\} \quad (23)$$

where, S_{min} , and S_{max} represents the minimum possible seed and maximum possible seeds respectively. F_p represents the fitness of any plant. The standard deviation in each iteration can be evaluated with Eq. (24) as:

$$\sigma_{iter} = \frac{(iter_{max} - iter)^n}{iter_{max}^n} (\sigma_{initial} - \sigma_{final}) + \sigma_{final} \quad (24)$$

where, $iter_{max}$ = the maximum number of iterations, σ_{iter} = standard deviation of the current step, n = nonlinear modulation index.

3.2.1.5 Forest Optimization algorithm Forest Optimization Algorithm (FOA) is a newly developed metaheuristics optimization algorithm that takes inspiration from the course of action of limited trees in forests that can live for a very long period. In this optimization algorithm, various methods of seed dispersal (mainly local seed dispersion and long-distance seed dispersal) are simulated and named local seeding and global seeding (Ghaemi and Feizi-Derakhshi 2014). Firstly, initialization of trees is performed as each tree represents a potential solution to a problem. Each tree shows a variable value and has the “Age” of that tree. The “Age” of the tree is always set at ‘0’ for every newly created tree (due to local or global seeding). The age of trees increases by unity after each local seeding phase except for freshly created trees in the local seeding phase. This increasing number of trees in the forest is used as a restricting mechanism to limit the number of trees in the forest.

A tree can be represented by an arrow of length $1 \times (M_{var} + 1)$ by the following Eq. (25) as:

$$Tree = [Age, v_1, v_2, v_3, \dots, v_{M_{var}}] \quad (25)$$

where, “Age” is the age of the related tree and M_{var} is the problem’s dimension.

This operator is applied on age trees with an age ‘0’ in the forest. The age of all the existing trees in the forest increases by unity during this process. As the iteration continues, the trees with an age higher than 0 increases which do not take part in the local seeding phase. The number of seeds falling on land close to trees which then turn to their neighbors is considered as “Local Seeding Changes (LSC)”. Also new trees with age ‘0’ are added in this stage. Two parameters (“Life time” and “area limit”) are used to inhibit the population of trees in FOA. The “Life time” is a predefined parameter representing the highest age of a tree. When the tree reaches its “Life time”, it is excluded from the forest and gets included in the candidate population. The bigger value of this parameter causes the formation of very old trees in the forest that do not get involved in the local seeding stage. Conversely, the smaller value of this parameter causes the trees to get excluded earlier from the forest hence providing a good probability of local search. The global seeding phase seeks to simulate the seed distribution of trees in the forest. After every iteration of the above steps, the best tree is chosen based on fitness and age values. The age value of the best tree is reset to ‘0’ so it can perform local seeding in the next step. The stopping condition is reached if (i) A certain number of iterations are completed. (ii) The observed value of a fit function of the best tree remains the same in multiple iterations. (iii) A desired level of accuracy is achieved.

4 Data description and statistics

Recognizing the crucial role of the database, this section aims to present the fundamental principles for creating a reliable and capable database. The performance of models is directly linked to the quality and the quantity of the data (Arif et al. 2024; Javed et al. 2024a, b; Sapkota et al. 2024a, b). The database collection is done from established sources where the datasets have undergone all the preprocessing techniques and are rid of multi-collinearity. Given the aforementioned significance attributed to the database, a new database has been compiled for modeling the data, incorporating results from several literatures. This database includes 681 mixed designs, with 324 data points extracted from Al-Shamiri et al. (2019) and the remaining 357 data are collected from Aslam et al. (2020). Detailed statistics of these datasets are presented in Table 2. The statistics indicate that the range of CS spans from 37.5 to 91.3 MPa, enhancing modeling efficiency within this range. The standard deviation (STD) of the CS data is 9.81, elucidating the dispersion of the data from the mean value. This comprehensive approach ensures that the database is robust and reliable, providing a solid foundation for developing accurate predictive computational models.

Figure 3 shows a histogram plot to determine the data’s distribution pattern. The histograms count the number of data

Table 2 Summary of statistical details of employed parameters

Parameters	Symbol	Units	Category	Count	Mean	STD	Min	Max	Range	Coefficient of Variation (CV)
Cement	Cement	(kg/m ³)	Input	681	400.27	87.27	160	600	440	0.218
Coarse Aggregate	CA	(kg/m ³)	Input	681	878.49	82.07	500	1486	986	0.093
Fine Aggregate	FA	(kg/m ³)	Input	681	787.89	102.86	342	1135	793	0.130
Water	Water	(kg/m ³)	Input	681	171.87	12.71	132	302.08	170.08	0.073
Superplasticizer	SP	(kg/m ³)	Input	681	1.68	2.10	0	12	12	1.252
Compressive Strength	CS	MPa	Output	681	51.97	9.81	37.5	91.3	53.8	0.188

points within specific bins to display the data distribution graphically. A histogram's x-axis shows the range of values in the dataset, while the y-axis shows how often data points appear within each column. These diagrams are particularly useful as they define the range of parameter values for which the optimal computational model that will be trained and proposed will be valid. Specifically, for parameter values smaller than the minimum value or greater than the maximum value, the developed model will not be reliable. These diagrams also play a crucial role in identifying areas where the parameter values are insufficiently represented, indicating that the database needs to be updated. This issue will be thoroughly discussed in a subsequent section titled "*Discussions, Limitations, and Future Recommendations*." To further elucidate the importance of these diagrams and their role in the development of reliable computational models, it is essential to understand their functions and implications. The diagrams therefore serve as a critical tool in defining the range of parameter values for which the proposed computational model will be valid. By clearly delineating these ranges, researchers ensure that the model's predictions are reliable only within the specified boundaries. This helps in preventing the application of the model beyond its intended scope, where its reliability would be compromised. For parameter values outside the defined minimum and maximum limits, the model's reliability diminishes. This is crucial in practical applications where adhering to validated parameter ranges ensures the model's accuracy and effectiveness, underscoring the importance of operating within the validated data range to maintain model integrity. The diagrams also highlight areas where the parameter values are not adequately represented in the dataset. Identifying these gaps is essential for improving the database, as it indicates regions where additional data collection is necessary. This continuous updating of the database ensures that the model remains robust and capable of handling a wide range of scenarios. Addressing data insufficiencies by updating the database with new and relevant data points enhances the model's reliability, ensuring that the

model can accurately simulate and predict behaviors across a more comprehensive range of parameter values. The subsequent section titled "Discussions, Limitations, and Future Recommendations" will provide a thorough analysis of these issues. It will include a detailed examination of the diagrams, the identified data gaps, and the implications for the model's reliability. Such discussions are vital for understanding the model's limitations and for making informed recommendations for future research and data collection efforts. By addressing the limitations identified through these diagrams, researchers can outline clear directions for future research. This proactive approach ensures that the computational models are continuously improved and updated, maintaining their relevance and reliability over time. By emphasizing these aspects, the importance of these diagrams in the development and validation of computational models becomes evident. They are not only tools for defining valid parameter ranges but also for identifying areas needing further data collection, thereby enhancing the overall quality and reliability of the models.

Comparably, Fig. 4 shows the Pearson correlation coefficient (R) between input and output variables. The correlation coefficient is a statistical metric that ranges from -1 to 1 . A range of -1 represents the most substantial negative relationship, whereas a value of 1 shows the most extreme positive correlation. The study shows that Cement (0.62) has a higher correlation with the development of CS. Similarly, CA (-0.1) shows a negative correlation. Likewise, FA (-0.33) shows a mild negative correlation along with Water (-0.03), showing a negative relationship, stating a decrease in CS with such components. Likewise, the SP (0.54) shows a higher positive relationship in the development of HSC.

5 Experimental methodology

The requirements mentioned by the Indian specifications are followed by the cement, sand, and coarse aggregates

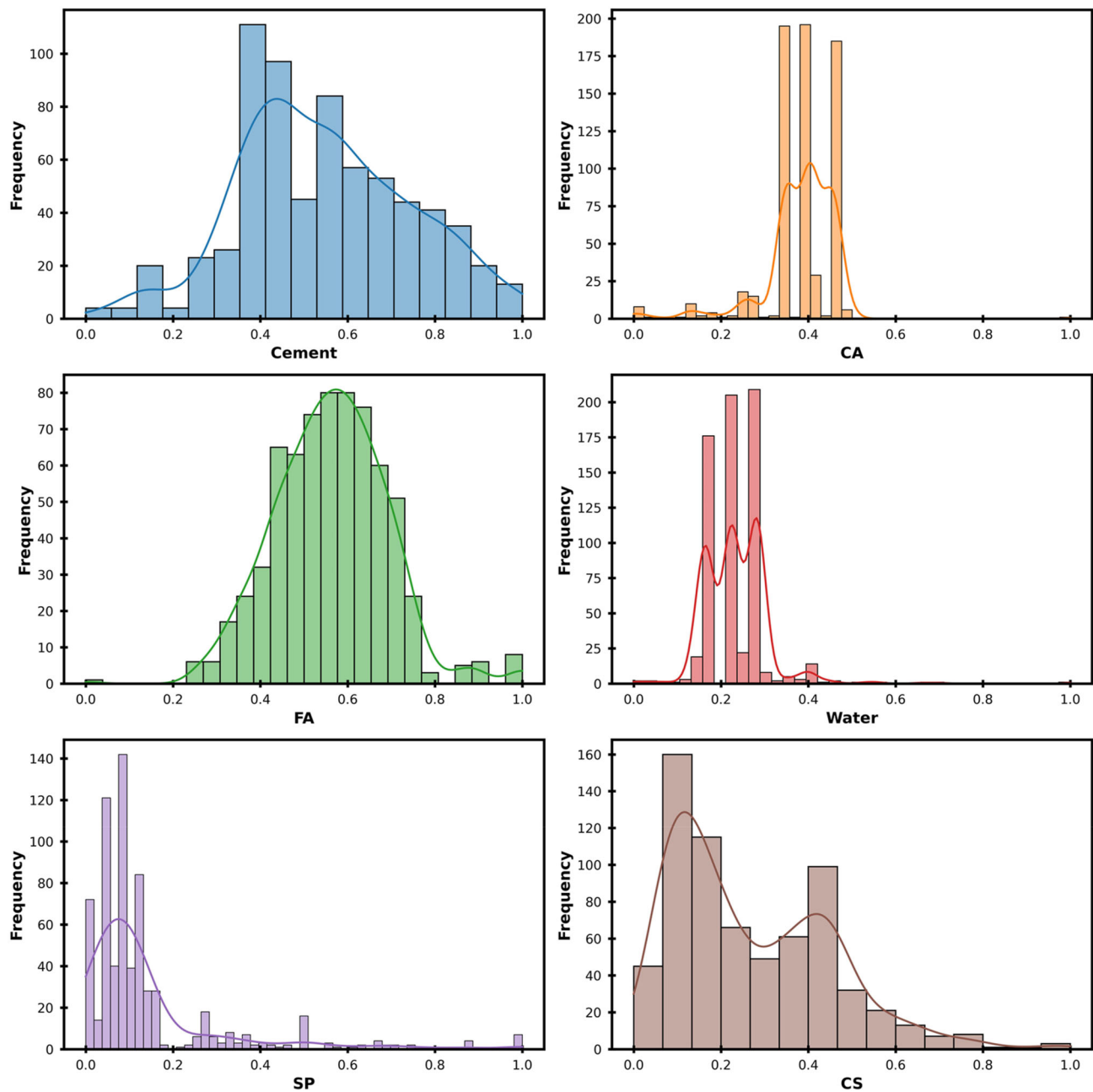


Fig. 3 Histogram plots with KDE lines

utilized during this study. OPC 53 grade was used in the research to conduct the experimental validation of High-strength concrete. The physical characteristics of the cement and its fine aggregates demonstrate that it meets the requirements declared in Indian standards of IS 12269:1989 and IS 383:2016. High-strength concrete is made by mixing cement, fine aggregates, coarse aggregate, water, and superplasticizers. An experiment that was conducted and illustrated in Fig. 5, was carried out to validate the current prediction methodology. Crushed stone fragments were used for the coarse aggregate, collected from a nearby hilly area, and river

sand was used for fine aggregates. The physical properties of cement, such as initial setting time, final setting time, fineness by sieving, specific gravity, and normal consistency, are 35 min, 517 min, 6%, and 31%, respectively. Furthermore, the physical properties of fine aggregates, such as dry density, water absorption, and specific gravity, are 1650 kg/m³, 1%, and 2.55, respectively. Additionally, the physical properties of coarse aggregate such as size, dry density, water absorption, and specific gravity are 20 mm, 1850 kg/m³, 2.7 and 0.73% respectively. Moisture content in both fine aggregate and coarse aggregate is considered negligible. A total

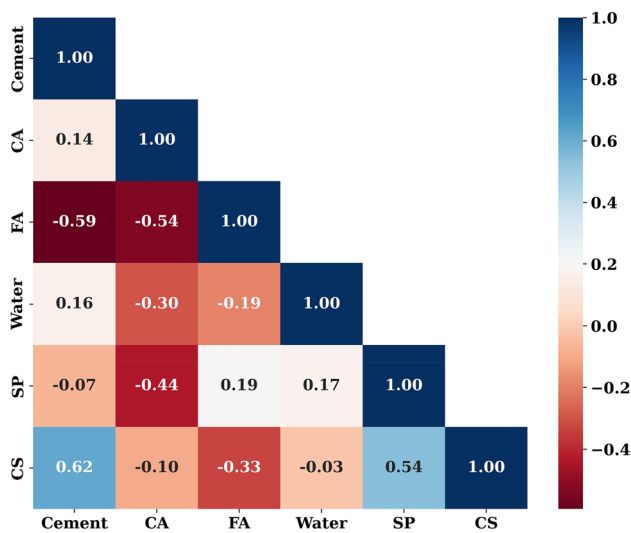


Fig. 4 Pearson correlation plot of employed features

of $10 \times 3 = 30$ cubes having a dimension of $150 \times 150 \times 150$ mm were cast to achieve the concrete's strength. After 24 h, the cubes were cast and then cured for 28 days in a curing tank under normal temperatures. The design mix carried out for M60 expected the strength of 60 MPa in 28 days was adopted for the study. Afterwards, the CS of the specimens was achieved in a standard compressive testing machine as shown as in Fig. 5.

6 Model building and hyperparameter tuning

The hybridization of the ML model has been proven to be one of the most efficient strategies for the effective prediction of CS. In the present study, K-fold cross-validation is used to evaluate the accuracy of ML models. The model is tested on the subgroup that has not been used for training after training on k-1 of the k subsets of the data (Ling et al. 2019). k iterations are carried out, and the validation data is used once in each iteration. Cross-validation offers

Fig. 5 Experimental Validation
a Materials preparation **b** Mixing
c Curing **d** Compressive strength test



a more unbiased assessment of a model's performance than a single train/test split, as it incorporates all data for testing and training purposes. The average performance for all k iterations is utilized to evaluate the performance assessment. Overfitting, which occurs in models exhibiting high performance on training data, can be mitigated using K -fold cross-validation (Ghojogh and Crowley 2019). The current study initially splits the training and testing datasets in a ratio of 80:20. The testing datasets, accounting for 20% of the total datasets, are maintained separately from the training datasets. The training datasets, consisting of 80% of the total datasets, are partitioned using tenfold cross-validation. This mitigates the bias in the training and testing sets.

The data distribution is split into ten folds, where training data accounts for nine folds and the remaining fold is used for testing/validation purposes (Mazumder et al. 2024; Lyngdoh et al. 2022). The testing fold position varies in each divide, and the model is developed by calculating the average of errors observed in each fold. The ultimate training model is determined by calculating the average of every model. Multiple combinations of attributes and traits are examined to determine the most effectively altered hyperparameters. Hyperparameters significantly impact the learning process and are crucial in establishing the model parameters during training. Hyperparameter optimization is essential to improve the functioning of ensemble approaches before training.

The hybridizations of tree-based ensemble and meta-heuristic algorithms help to mitigate the uncertainties and complex relationships among the features used. In the current study, six newly developed MHA are employed to optimize the hyperparameters of XGB to estimate the strength of the concrete. The parameters of XGB, like learning_rate, max_depth, and n_estimators, should be optimized to make the model robust and accurate. The optimization of such hyperparameters is done by using the MHA where exploration and exploitation based on the behavior is carried out. Before maximizing the hyperparameters of applied models, population size (n_{pop}), maximum number of iterations used (max_iter), lower and upper bound, and total hyperparameters to optimize (dim), are required to set up. Likewise, the optimal values of hyperparameters are determined by calculating the mean square error (MSE) in all the cases. In this study, the common MHA parameters are set as $dim = 3$; $max_iter = 100$; $n_{pop} = 50$. However, in cuckoo search, the probability of replacing nest's egg (P_a) = 0.25, and the scaling factors for levy flight (α) = 0.01. Similarly, the step size for water strider movement (β) = 0.1. Likewise, in leopard seal, the probability of shifting to the best individual (p_{move}) = 0.7 and the step size for movement of leopard seal (p_{seal}) = 0.1. Similarly, in Harris Hawk, the Harris Hawk exploration and exploitation constant is $r_1 = r_2 = 2$. Likewise, in forest optimization, the number of trees to

replace in each regeneration steps (n_{tree}) = 10. All these parameters are essential to adjust throughout the hybridization phase to optimize the hyperparameters of XGB and are presented in Table 3.

7 Performance metrics

Assessing the model's performance in both the training and testing sets is crucial. The model's performance on new, unknown data can be determined by using a performance indicator. These indicators enhance the model's generalization ability and robustness (Bardhan et al. 2023; Umar et al. 2021). Equations (26–30) provide the five distinct statistical parameters used in the present study. A more in-depth assessment that provides insights into the model's performance may be conducted using several statistical indicators.

$$R^2 = \frac{\sum_{j=1}^m (d_j - d_{average})^2 - \sum_{j=1}^m (d_j - y_j)^2}{\sum_{j=1}^m (d_j - d_{average})^2} \quad (26)$$

$$RMSE = \sqrt{\frac{\sum_{j=1}^m (d_j - y_j)^2}{M}} \quad (27)$$

$$MAE = \frac{\sum_{j=1}^m |(y_j - d_j)|}{M} \quad (28)$$

$$WI = 1 - \left[\frac{\sum_{j=1}^M (d_j - y_j)}{\sum_{j=1}^M (|y_j - d_{average}| + |d_j - d_{average}|)^2} \right] \quad (29)$$

$$VAF(\%) = \left(1 - \frac{var(d_j - y_j)}{var(d_j)} \right) \times 100 \quad (30)$$

where, d_j = measured value in j^{th} reading, y_j = predicted value in j^{th} reading, $d_{average}$ = mean of the measured value, M = total number of readings, VAF = Variance Account Factor.

8 Result and discussions

8.1 Performance of employed model

This study uses six hybridized algorithms, namely CSA-XGB, WS-XGB, LS-XGB, HH-XGB, IW-XGB, and FO-XGB, to predict the strength of HSC. The results of training and testing sets are tabulated in Table 4. From Table 4, it is evident that the CSA-XGB, WS-XGB, LS-XGB, IW-XGB, and FO-XGB all performed well. However, LS-XGB with $R^2 = 0.9829$, $RMSE = 1.3277$, $MAE = 0.5841$, $WI = 0.9955$, $VAF(\%) = 98.2929$, makes the model better performing

Table 3 Details of hyperparameters using the study

Model Names	Hyperparameters details	Parameters of MHA	Optimal value
CSA-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; P _a = 0.25; alpha = 0.01; dim = 3	0.478
	max_depth (5 to 50)		9
	n_estimators (10 to 200)		85
WS-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; beta = 0.1; dim = 3	0.448
	max_depth (5 to 50)		7
	n_estimators (10 to 200)		162
LS-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; p_move = 0.7; p_seal = 0.1; dim = 3	0.437
	max_depth (5 to 50)		6
	n_estimators (10 to 200)		150
HH-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; dim = 3; r ₁ , r ₂ = 2	0.119
	max_depth (5 to 50)		6
	n_estimators (10 to 200)		128
IW-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; dim = 3	0.531
	max_depth (5 to 50)		5
	n_estimators (10 to 200)		192
FO-XGB	learning_rate (0.1 to 1)	n_pop = 50; max_iter = 100; n_trees = 10; dim = 3	0.435
	max_depth (5 to 50)		6
	n_estimators (10 to 200)		88

Table 4 Performance of employed model on training and testing sets

Dataset	Indices	Ideal Range	CSA-XGB	WS-XGB	LS-XGB	HH-XGB	IW-XGB	FO-XGB
Training Set	R ²	1	0.998	0.998	0.998	0.996	0.998	0.998
	Rank Score		6	6	6	5	6	6
	RMSE	0	0.311	0.311	0.311	1.139	0.313	0.312
	Rank Score		6	6	6	3	4	5
	MAE	0	0.040	0.040	0.041	0.815	0.063	0.048
	Rank Score		6	6	5	2	3	4
	WI	1	0.999	0.999	0.999	0.996	0.999	0.999
	Rank Score		6	6	6	5	6	6
	VAF (%)	100	99.896	99.896	99.896	98.620	99.895	99.896
Testing Set	Rank Score		6	6	6	4	5	6
	R ²	1	0.9658	0.9761	0.9829	0.9714	0.9820	0.9810
	Rank Score		1	3	6	2	5	4
	RMSE	0	1.8773	1.5692	1.3277	1.7170	1.3618	1.4002
	Rank Score		1	3	6	2	5	4
	MAE	0	0.7840	0.6810	0.5841	1.1633	0.606	0.6123
	Rank Score		2	3	6	1	5	4
	WI	1	0.9909	0.9937	0.9955	0.9924	0.9952	0.9950
	Rank Score		1	3	6	2	5	4
	VAF (%)	100	96.5851	97.6138	98.2929	97.1434	98.2029	98.1002
	Rank Score		1	3	6	2	5	4
	Total Rank score		36	45	59	28	49	47

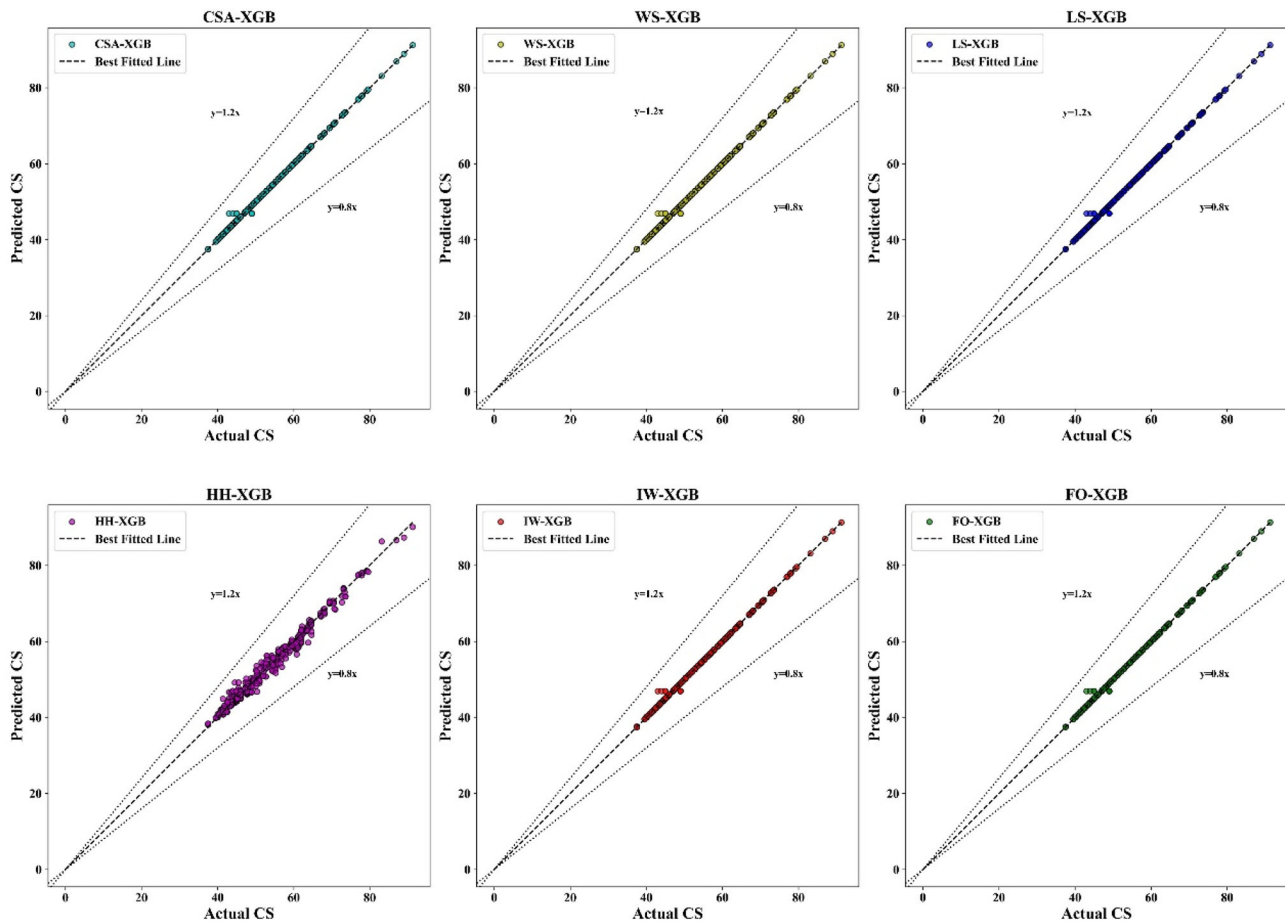


Fig. 6 Actual Vs Predicted Plot in Training Set

in testing sets. All the models performed equally well in both training and testing sets. However, among the adopted models, HH-XGB is the least-performing model, with $R^2 = 0.9714$, $RMSE = 1.7170$, $MAE 1.1633$, $WI = 0.9924$, $VAF (\%) = 97.1434$, and in testing sets. Similarly, Fig. 6 and Fig. 7 show the model's performances in training and testing sets, respectively. The error range of $\pm 20\%$ line in all algorithms reveals that the LS-XGB is the best among all the models.

It can be seen that all models consistently perform better in both the training and testing sets. Further, the rank analysis is based on the score; when the metrics performance is superior, the highest point, six, is given, and the lowest single point is given for the least performing metrics (Kumar et al. 2023). The summation of the total rank score in training and testing with the highest point is considered the consistent performing model across all the metrics. The total rank score of LS-XGB with 59 is regarded as the best model. However, the potential of IW-XGB is outstanding in both sets, with a score of 49. HH-XGB exhibits the lowest performance, with a score of 28, making it the least-performing model across all metrics.

8.2 Data visualization

The Taylor diagram is employed to assess and compare the performance of different models in predicting HSC's strength. This graphical representation employs radial axes to depict the correlation of predicted data and a circular grid to illustrate the standard deviation of observed data (Sapkota et al. 2024c). The anticipated results are presented as points along the radial axes, while the observed data is denoted by a point at the circle's center. The distance of each predicted point from the center corresponds to the standard deviation of the predicted data. In contrast, the angle formed by the radial axis and the line connecting the observed and anticipated points signifies the correlation between the two datasets. In the analysis of both the training and testing sets, the range of the SD line falls between points 9 and 10 on the Taylor diagram. The nearer line to the reference point is considered the best model. Figure 8a, b show the major plot details of all employed algorithms in the training and testing sets, respectively. All the models performed very well in both the training and testing sets. However, LS-XGB is regarded as the best

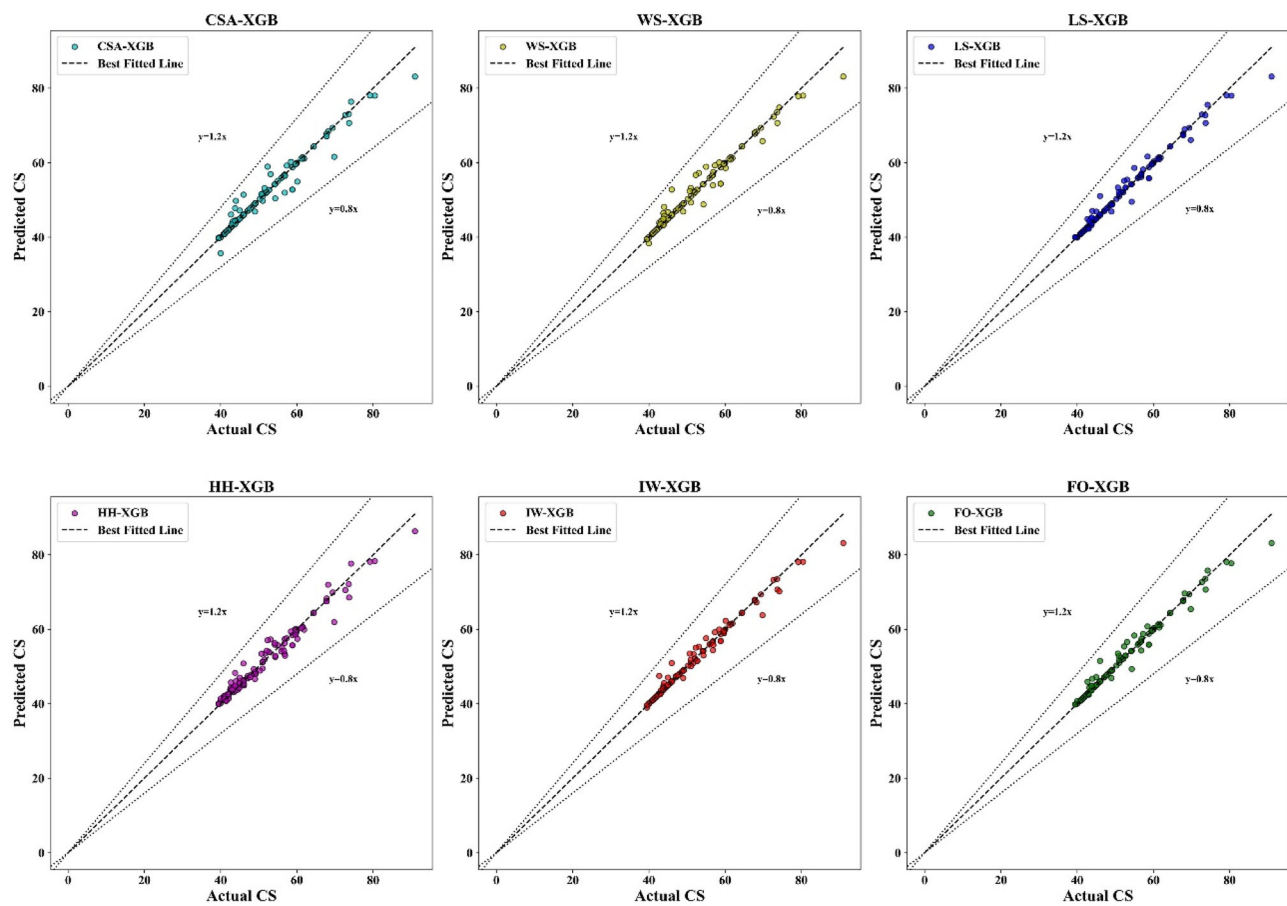
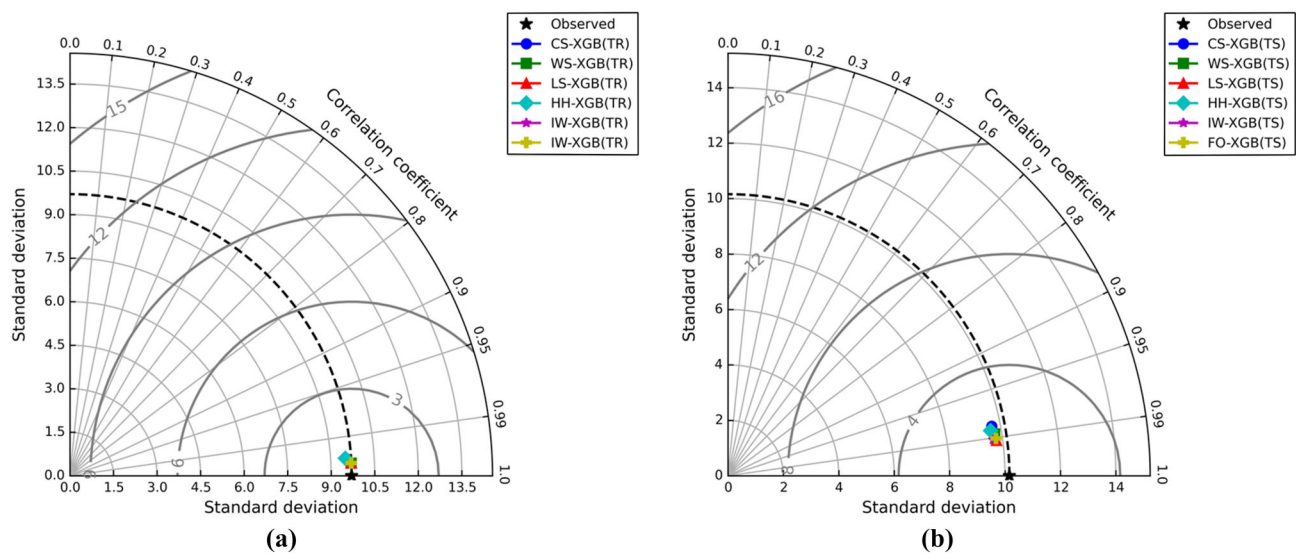


Fig. 7 Actual Vs Predicted Plot in Testing Set

Fig. 8 Taylor plot in **a** training set and **b** testing set

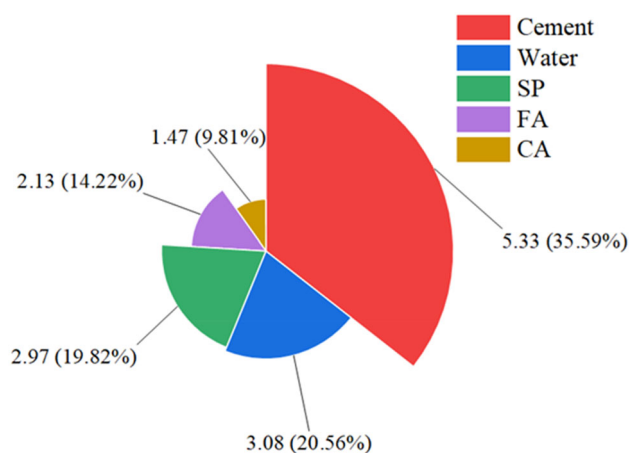


Fig. 9 Feature importance plots using LS-XGB

among all the models, with an SD range of 9.699 and 9.7764 in training and testing sets, respectively. The observed SD for train and testing sets is recorded as 9.7046 and 10.1589, respectively. This inference was drawn based on its proximity to the standard deviation line and the observed value. Correlation coefficients, closely aligned with the discerned values for the training and testing sets, further supported the conclusion.

8.3 SHAP analysis

Shapley Additives, also referred to as SHAP, is widely utilized in the field due to its ability to elucidate the behavior of models that ensemble machine learning methods struggle to comprehend. When estimating the contribution of each feature, Shap uses the Shapley values concept from game theory, which was first proposed by Lundberg and Lee (2017). The determination of the relevance of each feature's values is computed using Shapley values, which may vary in positive and negative based on the impact on the output (Covert et al. 2020).

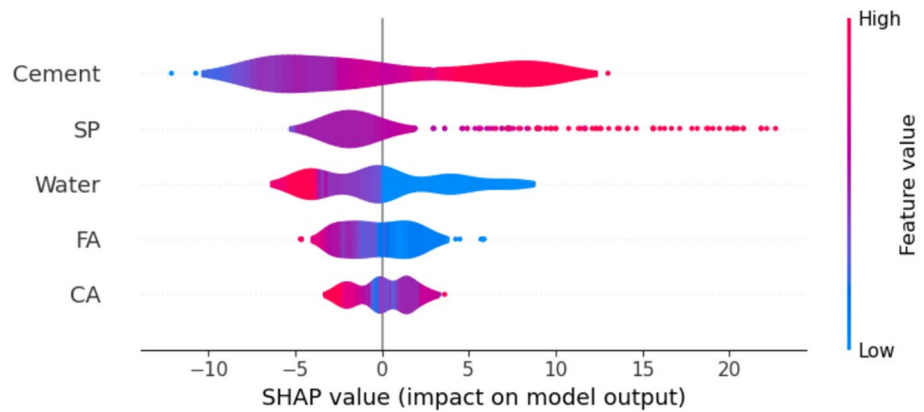
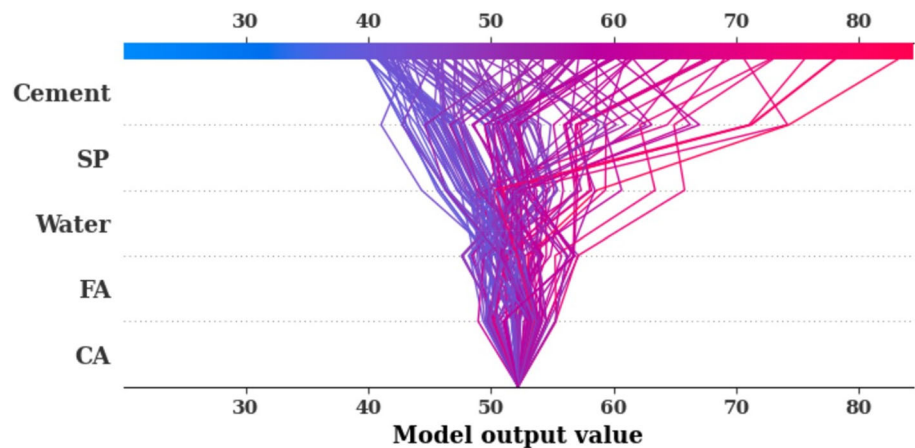
Using SHAP, a Feature importance plot helps identify the role of the best features in predicting CS, as shown in Fig. 9. They provide a feature importance Shapley score on the comparative evaluation of individual features that impact the output feature (Lundberg et al. 2020). In the plot, it is evident that Cement is the most influential feature with a weightage of 5.33 (35.59%) of the total weight, followed by water with a weightage of 3.08 (20.56%) and SP with 2.7 (19.82%) has similar weightage for the prediction of CS. Likewise, the FA feature with 2.13 (14.22%) and CA with the weightage of 1.47 (9.81%) are of the most minor importance. It is also surprising that the CA is slightly minor in importance, implying that the final algorithm cannot capture the insights for the final prediction of CS. The global interpretability of all the involved features is adequately captured.

Hence, the feature's importance improves the model explanation's behavior.

SHAP summary plots explain the behavior of the adopted model at the global level as shown in Fig. 10. The shift in Shapley values from blue (lower) to red (higher) pushes the model to a positive increase in the output variables (S. M. Lundberg et al. n.d.). Cement, being the most essential feature, states that with the rise in cement value, the model has a positive shift. Similarly, SP and CA also help push the model forward, while the increase in the Water and FA content results in decreased CS.

The decision plot offers a method for visually representing the collective impact of several variables on numerous predictions in an accumulative way. This allows for an evaluation of how various characteristics gradually contribute to the model's decision-making process, starting with the initial values and progressing to the final output of the model, as shown on the x-axis (Lundberg et al. 2018). Every line in the datasets, shown in Fig. 11, represents each occurrence. The average forecast of the model serves as the benchmark for each SHAP value, similar to how the intercept of a linear model is used to assess the coefficients of the linear model. The connection is shown on the y-axis, representing the model's characteristics. Typically, the organization of these components adheres to a hierarchy based on the significance of the data being shown, with the most important information placed at the top. Each prediction in the dataset is shown by a line to a particular color, with red indicating greater values and blue indicating lower values. The line meets the x-axis at the place of the picture most distant forward, giving the precise value of its related forecast. The hue of the line is dictated by the magnitude of the estimates, resulting in a gradual color change. The SHAP values for each feature consist of the model's initial forecast, which steadily increases from the plot's baseline. The output of the model's estimate is visually represented using a graphical resource, emphasizing each feature's impact. This plot is developed using the LS-XGB model, which exhibits a significant pathway for making a model decision. An increase may be seen from smaller magnitudes to higher ones. The test data shows a noticeable pattern defined by alternating zigzag lines. This pattern indicates that variables such as Cement and SP positively affect CS. Nevertheless, it has been demonstrated that all of the features have some relationships to the transformation of the model, providing an output value of 53.65 MPa. The phenomena may be elucidated by recognizing that some characteristics exhibit notable efficacy within a specific range, but their efficiency diminishes when used beyond this threshold.

The feature interaction plots using the SHAP plot can effectively address the effects of input variables by assessing the Shapley values. The interaction plots are developed using the LS-XGB method, which performed better in training and

Fig. 10 SHAP Summary plots using LS-XGB**Fig. 11** SHAP Decision plots using LS-XGB

testing sets. The horizontal features correspond to the feature value, and the y-axis corresponds to the SHAP values of each input variable (Lundberg et al. 2019). The positive SHAP values signify that incorporating features improves the output's predictive capability. This suggests that a positive correlation exists between the feature values and the corresponding output values, such that an increase in feature values corresponds to a corresponding rise in output values, and vice versa. Conversely, negative shape values signify that specific features exhibit a negative correlation or a decreasing influence over the model's output prediction. This suggests that an increase in feature values corresponds to a decrease in output values or vice versa.

The graphical representation in Fig. 12 depicts the correlation between Cement and CS. As cement content increases, SHAP values sharply increase with an optimal range of 500 to 900 kg/m³ in the presence of water in the 180 kg/m³ range. Likewise, CA, with a range of 600 to 800 kg/m³, makes CS impacts positively, and beyond that range, it starts decreasing. In the same way, with the increase of FA, there is a noticeable decrease in the SHAP values, stating the decrease in CS and an optimal range recorded in 500 to 700 kg/m³. The observed reduction in strength is attributable to the increase in the water

content with an optimized range of 150 to 175 kg/m³. Similarly, a positive shap value is seen with the increase in the SP value range. A significant rise in shap values with a maximum optimal range is noticed in 8 to 10 kg/m³ implying a strong relationship in the prediction of CS.

8.4 External validation using the proposed model

The quality of machine learning models depends on their ability to demonstrate remarkable performance and better generalization ability in terms of data distribution, noise in the data, and the model's explainability. Despite several variations in the data, a reliable model maintains stable and accurate predictions. To assess the robustness of the model to predict CS, external experimental validation becomes crucial since the developed model can be validated in entirely new, unseen conditions that the model is unfamiliar with. This technique incorporates rigorous statistical evaluations to verify the model's reliability, even when working with minimal datasets.

The experimental dataset used for the external validation is done by adopting the developed LS-XGB model. The result states that the LS-XGB model performs well in the validation set with $R^2 = 0.9241$, RMSE = 1.494, MAE = 1.2661, VAF

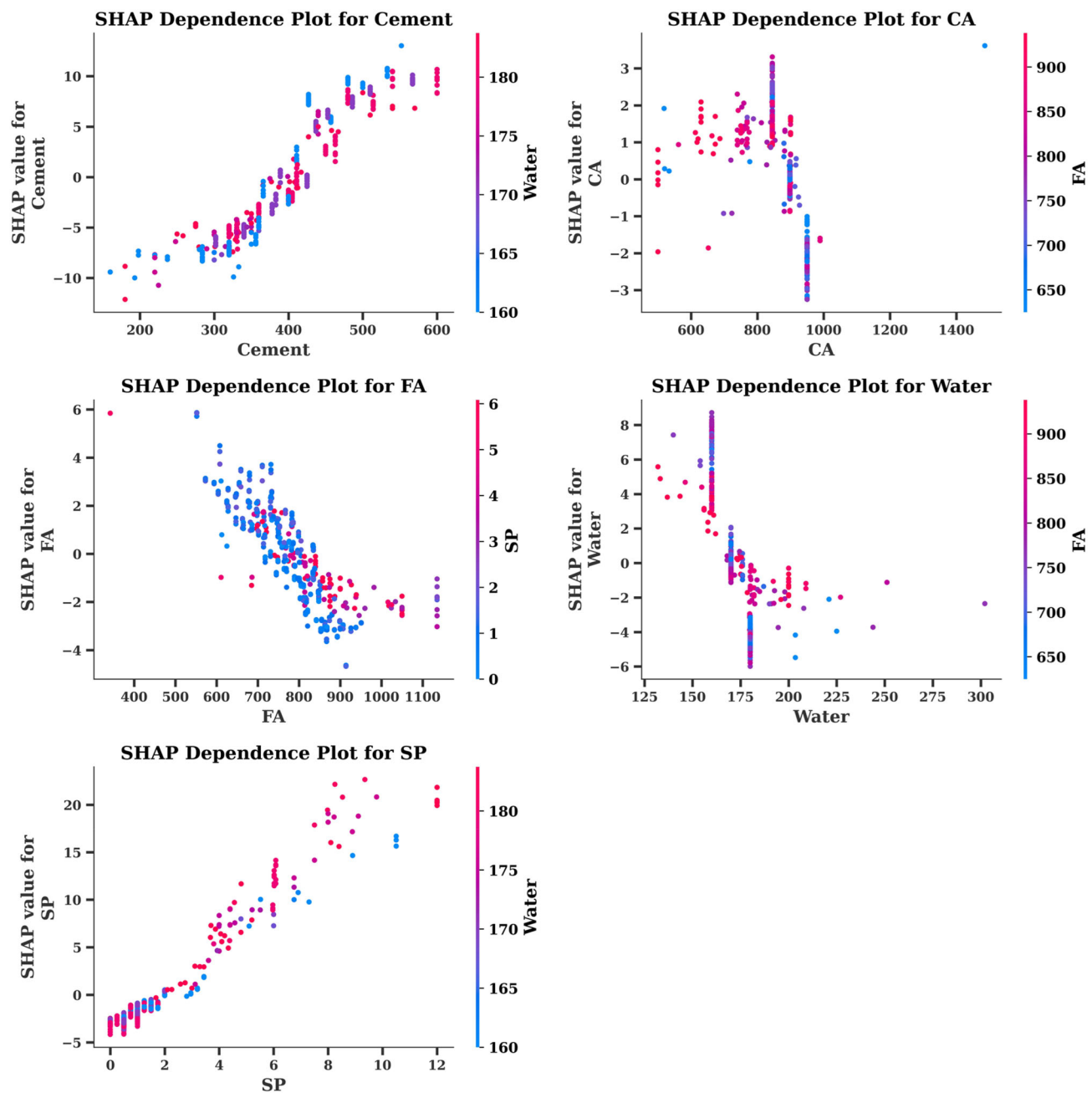


Fig. 12 Feature interaction plots using LS-XGB model

= 92.4%, and WI = 0.977. It confirms that the hybridized model LS-XGB is reliable and agrees well with the actual recorded data in the lab. Figure 13(a) shows the scattering of the experimental data for actual field data and lies closer to the best-fit lines, suggesting that most of the instances of the experimental data lie within the ($\pm$) 20 percent error. Likewise, Fig. 13(b) shows that in most of the index of experimental data, the actual line is overlapped with the predicted line by the LS-XGB model. However, in some data points, it is witnessed that there is a minimal distance line between

the actual and the predicted one, suggesting some minor error is present. Likewise, the error histogram plots stated in Fig. 13(c) show that the actual and the predicted errors are within ($\pm$ 3), confirming the minimal error exhibited by the model. The error frequency is concentrated within a difference of ($\pm$ 1) between actual and predicted data.

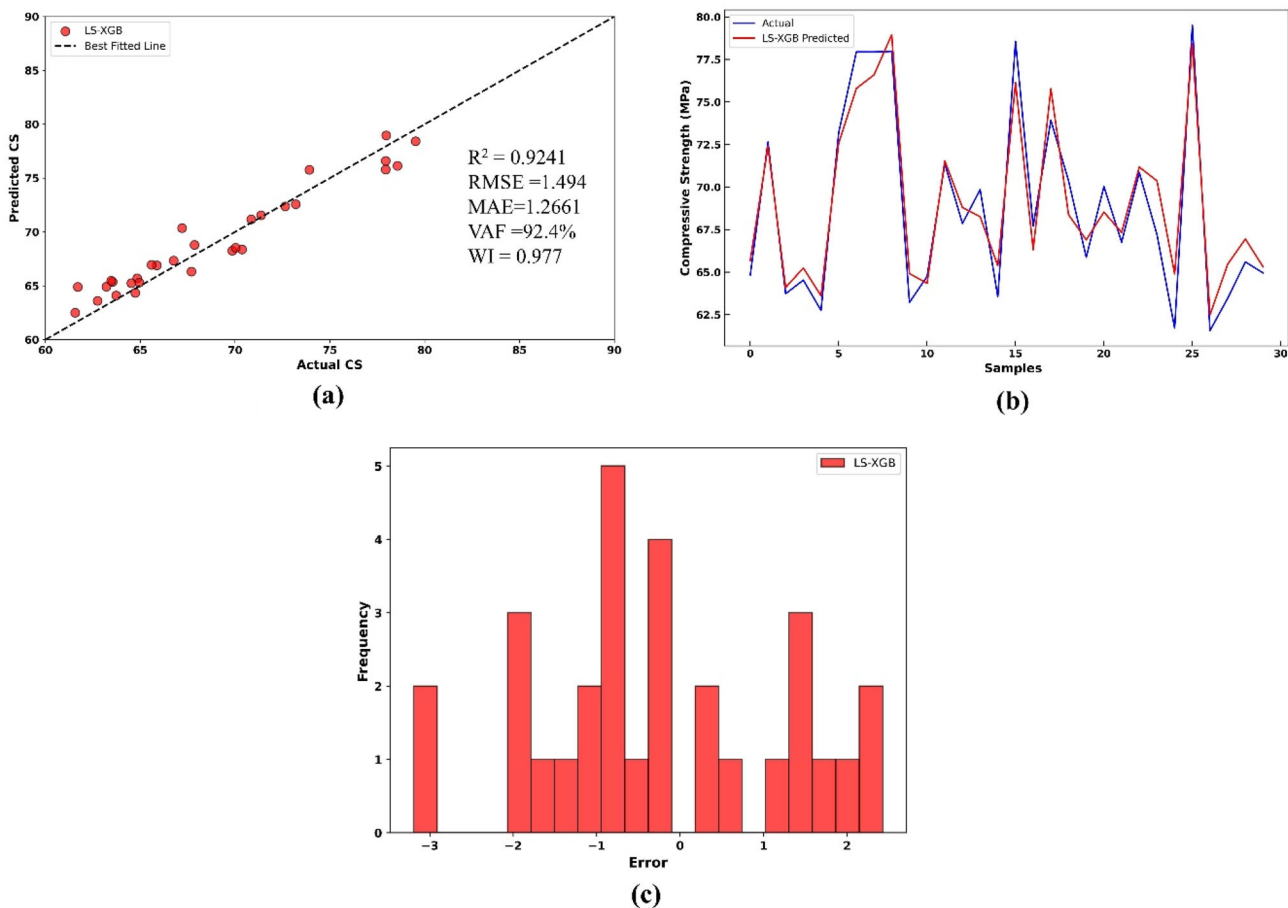


Fig. 13 Validation data sets **a** Scatter plot **b** Line plot **c** Error Histogram Plot

8.5 The Overfitting Problem

This subsection addresses the frequently encountered issue of overfitting during the development and training of predictive computational models. Overfitting is a pervasive problem that arises during the mathematical simulation of datasets (Ahmadi et al. 2024; Armaghani and Asteris 2021; Asteris et al. 2024; Cavaleri et al. 2022; Papanikolaou et al. 2024). It occurs when a model performs exceptionally well on the training data but fails to generalize to new, unseen data, leading to highly unusual predictions for other datasets. This highlights the importance of rigorously evaluating the overall performance of the proposed optimal LS-XGB model, particularly its predictive accuracy regarding the uniaxial CS of concrete relative to the water-to-binder ratio and various fine-to-coarse aggregate ratios.

To assess the robustness of the LS-XGB model, we computed the uniaxial CS of concrete across a wide range of parameters. The resulting graphs, presented in Fig. 14, demonstrate smooth curves that adhere to the expected trends. This consistency provides compelling evidence that

overfitting is not present in the developed and proposed optimal LS-XGB model.

Additionally, besides the smoothness of the curves, the lack of overfitting is evidenced by the fact that the testing datasets achieve nearly identical performance indicators to the training datasets (R^2 equal to 0.998 for the training datasets and 0.9829 for the testing datasets), as presented in Table 4. This high degree of similarity between the training and testing performance validates the reliability and generalizability of the LS-XGB model. The model's ability to maintain such high accuracy across different datasets underscores its effectiveness and robustness, making it a powerful tool for predicting the uniaxial CS of concrete in various conditions. It is to recognize that overfitting problems in computational predictive models typically arise when data are non-uniformly distributed, leading to regions with insufficient data representation. This imbalance can cause models to learn noise rather than the underlying data patterns, reducing their predictive validity on new datasets. Addressing these issues is vital for developing reliable and accurate predictive models that can be generalized across various conditions.

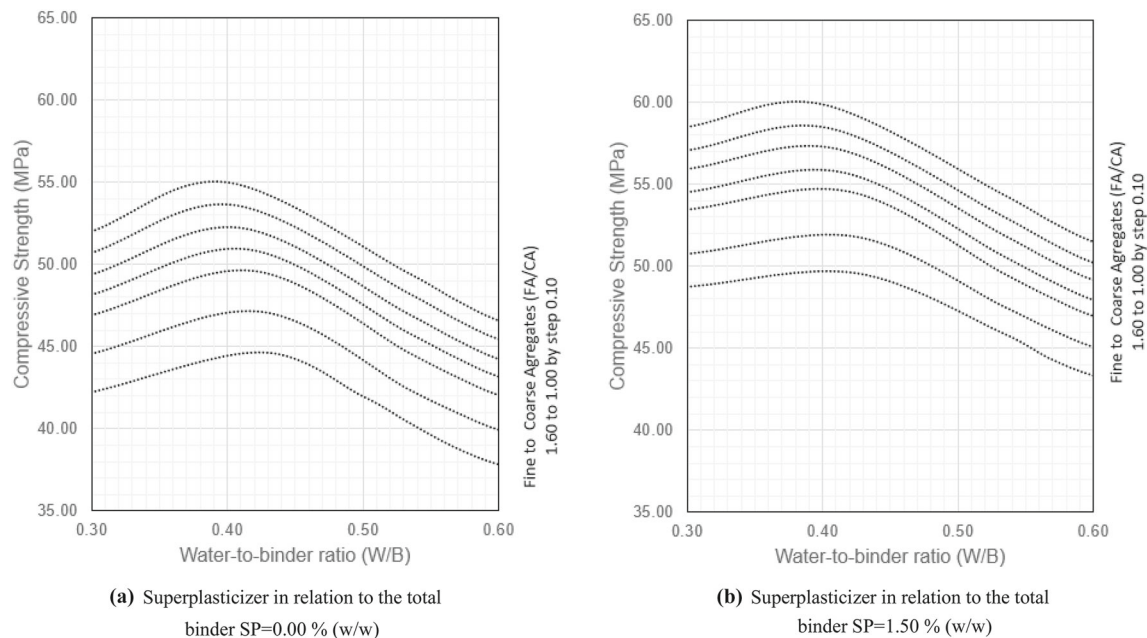


Fig. 14 Concrete CS vs water-to-binder ratio for superplasticizer percentage 0.00 and 1.50% (w/w) and values of Fine to Coarse Aggregates (FA/CA) from 1.60 to 1.00 by step 0.10 based on the optimum LS-XGB model

The figures presented in Fig. 14 not only indicate, through the smoothness of the curves, that overfitting is not occurring but also reveal the inherently nonlinear nature of concrete as a material. Specifically, they highlight the nonlinear dependency of uniaxial CS on the water-to-binder ratio, the fine-to-coarse aggregate ratio, and the superplasticizer content. These insights are critical for understanding the complex interactions within concrete mixtures and for optimizing material performance.

Despite the promising results, we did not proceed with designing additional diagrams, such as those depicting different superplasticizer ratios in relation to the total binder, due to the limited amount of data available for such specific values. This limitation emphasizes the need for a more comprehensive dataset to be used to study these relationships further. Addressing this limitation is a priority for the authors and will be pursued in future research. We plan to update the database accordingly and provide more detailed analyses in subsequent publications, thereby enhancing the model's robustness and applicability.

By expanding the dataset and refining our models, we aim to mitigate overfitting risks and improve the predictive accuracy of our computational models. This will enable us to better capture the intricate behaviors of concrete under various conditions, ultimately leading to more reliable and efficient designs in civil engineering applications.

8.6 Discussions, limitations and future recommendations

The presented hybridized meta-heuristic algorithms are evaluated in every possible way to estimate the CS of HSC in experimental settings. As mentioned in Sect. 8.1, the performance metrics, like R^2 , RMSE, MAE, WI, and VAF, are employed for assessment, confirming LS-XGB as a superior model. However, all other employed models performed remarkably well in both the training and testing sets, demonstrating the ability of hybridized models to predict. It is concluded that there has not been the presence of uncertainties, significant variations, or undesirable outcomes. Table 5 compares the current study with the existing literature and demonstrates that the adopted models are more robust. Overall, all the MHA, including the Cuckoo search algorithm, Harris hawks algorithm, Leopard seal algorithm, Forest optimization algorithm, Water strider algorithm, and Invasive weed algorithm, maintain a balanced approach in both exploration and exploitation phases. Furthermore, it explores the allocated hyperparameter space to find a global optimum value. Since hybridization is carried out with ensemble models, it is evident that the model will have complexities. However, tree-based ensembles are more explainable than neural networks. Thus, the hybridized ensemble models are more approachable and reliable for predicting the CS in real-time scenarios.

At this point, it is crucial to emphasize that any predictive computational model that results from appropriate design,

Table 5 Comparison of the current study with existing literature

Study reference	Type of concrete	Datasets	Models used	R ²	RMSE
Shen et al. (2022)	UHSC	372	Adaboost	0.82	13.1
			bagging	0.78	14.6
			XGB	0.9	7.6
Chen et al. (2022)	HSC	324	LSTM	0.997	0.508
			SVR	0.973	1.595
			CSA-XGB	0.9658	1.8773
Current Study	HSC	681	WS-XGB	0.9761	1.5692
			LS-XGB	0.9829	1.3277
			HH-XGB	0.9714	1.717
			IW-XGB	0.982	1.3618
			FO-XGB	0.981	1.4002

CSA Cuckoo Search, WS Water Strider, HH Harris Hawk, IW invasive Weed, FO Forest Optimization, LS Leopard Seal, SVR Support Vector Regression, LSTM Long short-term memory;

rigorous training, and thorough reliability testing can provide accurate and dependable predictions only for parameter values that fall within the specified range—between the minimum and maximum values for each of the input parameters. This means that the model's validity and accuracy are confined to these specified ranges, beyond which the reliability of the predictions cannot be assured.

For the LS-XGB model proposed in this paper, the boundaries that define its scope of application are presented in Table 6. These boundaries are established based on comprehensive data analysis and empirical validation, ensuring that the model performs optimally within these limits and provides precise predictions. For parameter values that lie outside these defined boundaries, the model's reliability is significantly compromised. This limitation arises because the model has not been trained or tested on data beyond these ranges, leading to increased uncertainty and potential inaccuracies in its predictions.

Emphasizing these boundaries is vital for practical applications of the LS-XGB model. Users must be aware that the model's predictions are only valid within the established parameter ranges, and caution should be exercised when interpreting results outside these limits. By adhering to the specified boundaries, the integrity and reliability of the model's predictions can be maintained, ensuring that it serves as a robust and dependable tool for forecasting and simulation.

Understanding and respecting these limitations is essential for the effective utilization of the LS-XGB model in real-world scenarios. Future work will involve expanding the dataset to cover a broader range of parameter values, thereby extending the model's applicability and reliability. Even though the current study uses compiled datasets for modeling and experimental datasets for validation, the increase in datasets can help to make the model more robust.

Several other newly developed hybridization techniques and multi-criteria analyses can be incorporated to make the study more realistic and capable of capturing further complexities of the data.

9 Conclusion

In this current study, advanced hybridized ensemble modeling techniques, namely, CSA-XGB, WS-XGB, LS-XGB, HH-XGB, IW-XGB, and FO-XGB, are successfully employed to estimate the 28th-day CS of HSC. The proposed model LS-XGB is subjected to 681 data collected from existing literature. The laboratory experimental data used in the validation subjected to the proposed models can be constructive for researchers and engineers to make a well-informed decision considering uncertainties. Some of the conclusions that can be drawn from this study are listed below: -

1. With 98% testing datasets and over 92% validation accuracy, the LS-XGB model proved to be the best at predicting. The predictive ability of this model with regard to HSC compressive strength is significant because scatter in the data is fairly minimal, and as a result, it is a relatively effective tool for practical use within established guidelines—this level of precision is very promising in relation to predicted outcomes.
2. A tenfold cross-validation method was used to guarantee not only the strength but also the generalizability of the models. Using this method not only reduced the risk of overfitting but also confirmed that the model predictions were generalizable across bins of the data.
3. The analysis of models established from HSC batch data established how specific HSC input sources are influential. Highlight from our analysis can be revealed that

Table 6 Statistical details of laboratory experimental data

Parameters	Symbol	Units	Category	Count	Mean	STD	Min	Max	Range	Coefficient of Variation (CV)
Cement	Cement	(kg/m ³)	Input	30	554.25	31.04	502.54	595.49	92.95	0.056
Coarse Aggregate	CA	(kg/m ³)	Input	30	901.35	28.55	845.97	946.45	100.53	0.032
Fine Aggregate	FA	(kg/m ³)	Input	30	810.69	54.13	706.10	892.72	186.62	0.0679
Water	Water	(kg/m ³)	Input	30	169.95	5.99	160.33	179.80	19.47	0.035
Superplasticizer	SP	(kg/m ³)	Input	30	6.73	3.11	1.05	11.84	10.79	0.47
Compressive Strength	CS	MPa	Output	30	68.81	5.42	61.57	79.52	17.95	0.08

cement and superplasticizers greatly affect CS of HSC. These results agree with experimental data and highlight the importance of these constituents as determining factors of concrete compressive strength.

- To interpret the model's predictions and determine the ideal ranges of important input features for HSC, a SHAP-based analysis was performed. The cement range was between 500 and 900 kg/m³, coarse aggregate (CA): 600 and 800 kg/m³, fine aggregate (FA): 500 and 700 kg/m³, water: 150 and 175 kg/m³, mass of superplasticizer (SP) was: 8 and 10 kg/m³. These ranges are quite consistent with the laboratory findings and serve as practical ranges of guidance for mix design of HSC.
- The credibility of the proposed models was further established through experimental validation. The experimental results were consistent with the model predictions, confirming the ability of the LS-XGB model to be used in real-world problems. The clustering of experimental and predicted data confirms the validity of the model, and its utility in optimizing concrete mix designs in practice.

The LS-XGB hybridized model demonstrates a strong and reliable method for predicting HSC compressive strength. This research presents an important pathway for refining concrete mix design whilst reconciling the parameters of matter at hand through the combination of data-centric and experimental approaches. Further studies should be performed to cover the hybridized ensemble model in civil engineering domains to improve predictive capacity and decisions in engineering problems.

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draft; Alexey N. Beskopylny: Data curation, Formal analysis, Validation, Visualization, Writing—original draft; Sandesh Dahal: Data curation, Formal analysis, Validation, Visualization, Writing—original draft; P. G. Asteris: Conceptualization, Investigation, Methodology, Software, Supervision, Writing—original draft, Writing—review & editing.

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Data availability statement The data used in the analysis is provided with the reasonable request from corresponding author.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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